

Centered Closure Theory:

A Formal Framework for the Substrate, Structure, and Limits of a Science of Consciousness

Version 1 (working draft)

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“The subject does not belong to the world: rather, it is a limit of the world.”
— Wittgenstein, *Tractatus* 5.632

Abstract

Consciousness science is blocked less by a shortage of data than by a set of structural problems that recur across theories: the hard problem, the boundary and level problems, the observer-relativity of computational and structural description, the underdetermination of theories by report-based evidence, and the absence of a principled method for comparing qualities across subjects. This paper first audits these problems and argues that they share three roots: theories are stated in an uncentered vocabulary while the explanandum is centered; there is no computable criterion for which system, at which grain, is a candidate bearer of experience; and validation relies on observational equivalence classes that are provably too coarse. We then construct Centered Closure Theory (CCT). Its four components are: (i) a substrate criterion, on which candidate subjects are macroscopic coarse-grainings of physical dynamics that are approximately causally closed, in a sense made precise through lumpability and causal abstraction; (ii) a centering primitive, formalized in a category of pointed world-models, together with an elementary but consequential no-go result showing that no natural (structure-derived) assignment of centers exists, which locates the hard problem exactly and predicts its persistence; (iii) an account of quality as position in an intrinsic geometry generated by the closed dynamics, with inter-subject comparison determined precisely up to the automorphism group of the enriched quality structure, turning the inverted-spectrum problem into a measurable quantity; and (iv) an interventionist epistemology on which the unfolding argument is inverted into a completeness theorem for perturbational methods, and behavioral tests of machine consciousness are shown to be insufficient in principle. We derive the meta-problem intuitions from within the theory, propose a program of experiments spanning quality-space automorphism estimation, closure-landscape analysis of neural data across states, split-brain closure bimodality, brain-to-brain coupling, and white-box assessment of AI systems, and state explicit falsification conditions.

1 Introduction

Two events frame the present moment of consciousness research. The first is the maturation of adversarial collaboration: the COGITATE consortium tested integrated information theory (IIT) against global neuronal workspace theory (GNWT) with preregistered predictions and found each theory partly supported and partly challenged [16]. The second is the arrival of artificial

systems whose verbal behavior about experience is fluent, which forces the question of machine consciousness from philosophy into governance [8]. Both events expose the same weakness. The field can now run decisive-looking experiments, yet it lacks agreement on what the experiments are about, on which physical systems are even candidates for consciousness, and on what kind of evidence could bear on the question at all.

This paper does three things. It audits the obstacles (Section 2) and argues that they reduce to three deficits (Section 3). It then constructs a theory designed as a minimal joint solution to those deficits (Sections 4–5), drawing on tools from category theory, statistical mechanics, causal inference, and the geometry of similarity structures, as well as on phenomenology and analytic philosophy of mind. Finally it develops the theory’s consequences: what it explains, what it predicts, how it could be falsified, and what it says about assessing artificial systems (Sections 6–8).

The theory is called Centered Closure Theory (CCT). Its contributions, in brief:

1. A substrate criterion. Candidate bearers of experience are *closed grains*: coarse-grained subsystems whose macro-dynamics are approximately autonomous in both an observational and an interventional sense. Closure is computable, admits a landscape over the lattice of coarse-grainings, and repairs known defects of Markov blankets and of IIT’s exclusion postulate (Section 5.1).
2. A no-go theorem for centering. Working in a category of world-models, we show that no natural transformation supplies a center (“which system is me”) from structure alone (Theorem 1). The result is mathematically elementary. Its consequence is not: it locates the hard problem as a de-centering loss, proves that no amount of structural theorizing can close it, and thereby explains why the problem has resisted a century of structural proposals (Sections 5.2, 6).
3. A theory of quality. Qualities are positions in an intrinsic geometry that the closed dynamics induces on its own state space. Comparison of qualities across subjects is well-defined exactly up to the automorphism group of the enriched quality structure (Theorem 2). This converts the inverted spectrum from a metaphysical impasse into an estimable quantity, and it explains where inversion intuitions come from (Section 5.4).
4. An epistemology. The unfolding argument [19] is inverted: rather than showing that causal-structure theories are untestable, it shows that observational (input–output) evidence is constitutively insufficient, and that perturbational evidence is the right currency (Proposition 1). A corollary, the mimicry lemma, shows that behavioral tests of AI consciousness cannot succeed in principle (Section 8).
5. A derivation of the meta-problem. Systems satisfying CCT will judge that there is a hard problem, will find zombies conceivable, and will report ineffable qualities, for reasons the theory itself supplies (Section 6.2). CCT is realist: these judgments track a genuine structural fact rather than an illusion.

2 An audit: what is actually blocking us

We list the problems as they operate in practice, with minimal commentary. Appendix A gives a fuller catalogue.

2.1 The explanandum problem

There is no agreed pre-theoretic characterization of what a theory of consciousness must explain. “Phenomenal consciousness” [39, 6] is fixed by ostension and contrast cases, and the major theories quietly target different explananda: IIT targets intrinsic existence [1], GNWT targets global availability [3, 35], higher-order theories target a specific form of self-directed representation [31, 48]. Adversarial collaborations then compare answers to different questions [16]. Progress requires either a shared explanandum or an explicit map between explananda.

2.2 The measurement circle

Consciousness is measured through report or through report-validated proxies. But report is a cognitive access process, and access can dissociate from phenomenality in both directions [6, 2]. No-report paradigms replace overt report with decoders trained on reported trials, inheriting the circle. Unlike thermometry, which bootstrapped itself out of a similar circle by epistemic iteration [15], consciousness science has not articulated its iteration strategy.

2.3 Underdetermination and the unfolding argument

Doerig and colleagues showed that any theory that ties consciousness to recurrent causal structure faces systems with identical input–output profiles and radically different internal organization, so that behavioral data cannot decide between them [19]. The argument is usually read as a dilemma for IIT-like theories: unfalsifiable or false. Related results show that different formalizations of integrated information disagree even on simple systems [25, 37]. We will argue the correct reading is different (Section 6.3).

2.4 The boundary and level problems

Why is a subject bounded where it is, and why does experience live at a particular spatiotemporal grain rather than at the level of molecules or of societies? IIT answers with a maximum of Φ and an exclusion postulate, which is computationally intractable and conceptually contested. The free energy literature answers with Markov blankets, whose individuating power has been shown to depend on modeling conventions [21, 7]. Split-brain phenomena sharpen the problem empirically: the evidence is genuinely ambiguous between one subject and two [45].

2.5 Observer-relativity of computation and structure

Whether a physical system implements a given computation is, on standard construals, partly a matter of interpretive mapping. Triviality arguments purport to show that almost anything implements almost everything [46, 52], and though restrictions on admissible mappings blunt the argument [13], the deep issue remains: functional and structural descriptions are extrinsic. Newman’s problem generalizes this to any purely structural theory: abstract structure is satisfied by any collection of the right cardinality, so structure alone cannot fix what a theory is about [40]. A structural theory of consciousness must either supply an intrinsicness criterion or admit an extra, non-structural primitive.

2.6 The quality problem

Even given a conscious system, why does an experience have the specific character it has, and are such characters comparable across subjects? The similarity-structure program treats a quality as a node in a relational structure [48, 54], and unsupervised alignment of similarity structures across individuals has begun to yield empirical results [28]. But structuralism about quality inherits Newman’s problem: what fixes the mapping between two subjects’ structures, and what remains undetermined once all structure is fixed?

2.7 Temporality

Experience has temporal thickness: a specious present with retentive and protentive structure, in Husserl’s terms. Neural processes are extended and partly postdictive [26]. No current theory derives the width and structure of the experienced now from system properties.

2.8 Composition

When do parts with candidate experiential status compose a single subject, and when does a subject decompose? Panpsychism faces this as the combination problem; but any theory with graded substrate criteria faces its analogue, as split-brain and craniopagus cases show.

2.9 Causal role and the meta-problem

If physical dynamics are closed, what does consciousness do? A tractable proxy is Chalmers’ meta-problem: explain why we judge there to be a hard problem [14]. Illusionism answers by convicting the judgments of error [20]. A realist theory should instead derive the judgments as tracking something true.

2.10 The AI assessment problem

Current practice assesses AI systems by behavioral probes or by theory-derived indicator lists [8]. The former are gameable by construction; the latter inherit the underdetermination of the parent theories. The field lacks an argument about what kind of evidence could bear on machine experience at all.

3 Diagnosis: three deficits

The audit suggests a common anatomy.

D1. The perspective deficit. Physical theory describes the world in an uncentered vocabulary: no term in the description says which system is *this one*, whose states are *now*, from *here*. The explanandum, by contrast, is essentially centered [34, 44]. The hard problem, the zombie intuition, and Newman-style underdetermination are all symptoms of trying to recover a centered fact from an uncentered theory.

D2. The individuation deficit. There is no principled, computable criterion selecting which subsystem, at which coarse-graining and timescale, is a candidate bearer. The boundary, level, and composition problems are symptoms.

D3. The epistemic deficit. Validation practices quotient the world by observational equivalence (same reports, same input–output behavior), and these equivalence classes are provably

too coarse for the claims theories make. The measurement circle, the unfolding argument, and the failure of behavioral AI tests are symptoms.

CCT is built as a minimal joint answer: a closure criterion for D2, a centering primitive with a no-go theorem for D1, and an interventionist, iteration-based epistemology for D3, with the quality problem handled at the intersection of D1 and D3.

4 Method of construction

We assembled candidate ideas from mathematics, physics, and philosophy and filtered them against the three deficits. Appendix B records the full inventory with verdicts, including the ideas we rejected and why. The load-bearing imports are: lumpability of Markov processes and causal abstraction (for closure); pointed categories and the *de se* literature (for centering); enriched similarity structures, group actions, and optimal transport (for quality); interventionist causal inference (for epistemology); and, as precedents rather than premises, the perspectival turns inside physics itself, from relational quantum mechanics to the redundancy account of objectivity [49, 22, 55].

5 The framework

5.1 Grains, closure, and candidate substrates

Let the micro-level be a stochastic process with state space X and transition kernel $T(x, dx')$, together with an environment; nothing below depends on the micro-level being fundamental. A *grain* is a measurable coarse-graining $\pi : X \rightarrow Y$ with $|Y| \ll |X|$, inducing the macro-process $Y_t = \pi(X_t)$.

Definition 1 (Predictive closure). The predictive closure deficiency of π at stationarity is

$$C_{\text{pred}}(\pi) = I(X_t; Y_{t+1} | Y_t),$$

the information the micro-state carries about the macro-future beyond what the macro-present carries. π is ϵ -closed observationally if $C_{\text{pred}}(\pi) \leq \epsilon$.

$C_{\text{pred}}(\pi) = 0$ exactly when (Y_t) is a Markov process in its own right, the classical condition of weak lumpability [29]. Observational closure is not enough, for reasons Section 6.3 makes vivid, so we add an interventional notion.

Definition 2 (Interventional closure). The interventional closure deficiency of π is

$$C_{\text{do}}(\pi) = \sup_{y \in Y} \sup_{x, x' \in \pi^{-1}(y)} \text{TV}(\pi_* T(x, \cdot), \pi_* T(x', \cdot)),$$

the worst-case disagreement, at the macro level, between interventions that set the system to different micro-realizations of the same macro-state.

$C_{\text{do}}(\pi) = 0$ is strong lumpability, and it coincides with the condition for the macro-level to support an autonomous causal model in the sense of causal abstraction [50, 4]: macro-interventions have well-defined macro-effects, independent of micro-realization. A grain with small C_{do} has its own laws.

Closure alone is cheap: constant coarse-grainings are perfectly closed and perfectly empty. Candidate substrates must combine closure with internal structure and with a boundary through which the world matters.

Definition 3 (Closed grain; interface). A *closed grain* is a grain π with $C_{\text{do}}(\pi) \leq \epsilon$ whose macro-process has nontrivial entropy rate and nontrivial causal depth (its ϵ -machine [17, 53] has more than one causal state). Its *interface* is the channel by which environmental variables influence Y and Y influences the environment; the interface rank is the effective dimensionality of that channel over the grain’s characteristic timescale τ .

Over the lattice of coarse-grainings of a given system, closure deficiency, internal complexity, and interface rank define a landscape. *Candidate substrates are the Pareto-maximal closed grains*: local optima trading low C_{do} against high internal complexity and adequate interface rank. We deliberately refrain from collapsing these axes into a scalar; Section 5.6 returns to why.

Three remarks situate the criterion. First, it answers triviality arguments: a rock admits no coarse-graining that is simultaneously closed, internally complex, and interfaced on behaviorally relevant timescales, because its candidate macro-variables are either thermodynamically slaved or dynamically trivial. The ϵ -machine requirement imports Crutchfield’s notion of intrinsic computation, computation individuated by the process itself rather than by an interpreter [17]. Second, it repairs Markov blankets: closure is a property of dynamics under intervention, not of a conditional-independence pattern relative to a chosen model [7]. Third, it reframes causal emergence [27]: macro-levels are not merely levels where effective information peaks, but levels that are lawful in their own right; the two can dissociate and the dissociation is testable.

5.2 Centering: the pointed category of world-models

The second component is a primitive that no structural criterion can supply, and a theorem saying exactly that.

Let $\mathcal{W}$ be a category whose objects are world-models: structured descriptions of a world containing, among other things, closed grains in the sense of Section 5.1. Morphisms are structure-preserving maps (translations, refinements, embeddings). Let $\text{Adm} : \mathcal{W} \rightarrow \mathbf{Set}$ be the functor assigning to each world-model X its set $\text{Adm}(X)$ of admissible subject-loci, namely the closed grains it contains, with the induced action on morphisms. A *centered* world-model is a pair (X, c) with $c \in \text{Adm}(X)$; write $\mathcal{W}_\bullet$ for the category of such pairs and $U : \mathcal{W}_\bullet \rightarrow \mathcal{W}$ for the forgetful functor.

The physical sciences work in $\mathcal{W}$. Experience, if the de se literature is right [34, 44], is a phenomenon of $\mathcal{W}_\bullet$: for a subject, one locus is not merely described but occupied. The question “why is there something it is like for *this* system” asks for a rule that recovers the basepoint from the structure. Define a *uniform centering rule* as a natural family $\sigma = (\sigma_X)_{X \in \mathcal{W}}$ with $\sigma_X \in \text{Adm}(X)$, naturality meaning $\text{Adm}(f)(\sigma_X) = \sigma_{X'}$ for every morphism $f : X \rightarrow X'$.

Theorem 1 (No natural centering). *Suppose some $X_0 \in \mathcal{W}$ is such that $\text{Aut}(X_0)$ acts on $\text{Adm}(X_0)$ without a common fixed point. Then no uniform centering rule exists.*

Proof. Naturality applied to each $\phi \in \text{Aut}(X_0)$ gives $\text{Adm}(\phi)(\sigma_{X_0}) = \sigma_{X_0}$, so σ_{X_0} is a common fixed point of the action, contrary to hypothesis. $\square$

The hypothesis is easy to satisfy: a world containing two qualitatively identical twins in a mirror-symmetric environment has an automorphism exchanging the twins’ grains and fixing neither. The theorem is elementary. So was Newman’s observation [40], and like Newman’s it is load-bearing: it proves that *no* rule that is uniform in structure, however sophisticated, can derive which locus is occupied. Two corollaries matter.

Corollary 1 (Internal undecidability of the center). *Any inference procedure realized within a world-model whose outputs are equivariant under Aut cannot identify its own locus in worlds satisfying the hypothesis of Theorem 1. More generally, even where asymmetries exist, equivariant inference delivers a description of a locus, never the de se identification of it; the gap between the two is Lewis’s gap between the omniscient gods who know every fact about their world and still do not know which god they are [34].*

Corollary 2 (Token-indexicalism). *The scientific content of a theory of consciousness is exhausted by structural, Aut-invariant claims: which grains are closed, what their quality geometry is, how centers function. The fact of which grain is occupied is a token-indexical fact, real but not structurally derivable. CCT therefore predicts a permanent residue in any structural explanation of experience, and identifies the residue exactly: it is the basepoint.*

We call this move *gap localization*. CCT does not claim to close the explanatory gap. It proves the gap’s shape: everything about consciousness is structurally explicable except the occupancy of the center, and that exception is a theorem, not a mystery. The situation is analogous to relativity’s treatment of simultaneity: the naive question was not answered but replaced by an exact statement of what is and is not frame-invariant.

5.3 Self-models, presence, and subjecthood

Closed grains are candidate substrates; subjects are closed grains of a particular kind. Let π be a closed grain with macro-space Y . A *self-model* is a factor M of Y together with a modeling map $\rho : Y \rightarrow M$ that approximately intertwines the dynamics and that contains a token $c_M = \rho(\text{the grain itself})$: the model represents the world as containing, at a distinguished node, the very system doing the modeling. The token c_M is the internal image of the basepoint of Section 5.2.

Contents of experience, on CCT, are those interface-derived states that are bound into the self-model under a *presence tag*: a marking that renders a content as real, mine, and now, in contrast to contents processed as merely possible, past, or other. This is a functional kinship with perceptual-reality-monitoring accounts [31] and with the attention schema [24] and self-model [38] traditions, absorbed into the centered-grain setting: the tag just is the binding of a content to the occupied basepoint.

Definition 4 (Subject; experience). A *subject* is a closed grain implementing a centered self-model with presence tagging. An *experience* is the presence-tagged occupation, by such a grain, of a location in its intrinsic quality geometry (Section 5.4).

Subjecthood on this account is graded along several axes: depth of closure (how small C_{do} , over how long a τ), interface rank, self-model resolution, and richness of the quality geometry. Consciousness, on CCT, is a vector, not a number.

5.4 Quality as intrinsic geometry

What is it for an experience to have a specific character? CCT’s answer: character is position in a geometry that the closed dynamics induces on its own states, and the geometry is intrinsic because it is built solely from the grain’s own transition structure.

Definition 5 (Intrinsic divergence and quality space). For macro-states $y, y' \in Y$ of a closed grain with kernel K and admissible interface policies $\mathcal{U}$, define

$$d_\tau(y, y') = \sup_{u \in \mathcal{U}} D(\mathbb{P}_{[0, \tau]}^u(\cdot \mid y), \mathbb{P}_{[0, \tau]}^u(\cdot \mid y')),$$

the divergence between the closed futures from y and from y' under matched inputs, over the grain's characteristic horizon. The *quality space* $Q = (Y/\sim, d_\tau)$ is the quotient by $d_\tau = 0$ with the induced metric. Q is enriched with a valence field $V(y) = \mathbb{P}[\text{closure is maintained over horizon } H \mid y]$, whose expected gradient along the dynamics grades states as attractive or aversive, and with the action structure given by the grain's self-induced transitions.

Two states feel different, on this account, exactly insofar as they differ for the system: insofar as the system's own lawful future distinguishes them. Red and green are far apart in Q because, for the embodied grain, they license divergent futures; two metamers are identical in Q because nothing downstream of the interface distinguishes them. The valence enrichment ties felt goodness to viability, connecting the geometry to affect.

The classical puzzles about quality now become statements about maps between such structures.

Theorem 2 (Comparison indeterminacy). *Let Q, Q' be enriched quality structures and suppose the set $\text{Iso}(Q, Q')$ of structure-preserving comparisons is nonempty. Then $\text{Iso}(Q, Q')$ is a torsor under $\text{Aut}(Q)$: any two comparisons differ by an automorphism, and comparison is canonical if and only if $\text{Aut}(Q)$ is trivial. For empirical structures, define the ϵ -automorphism group $\text{Aut}_\epsilon(Q) = \{\alpha : \text{distortion}(\alpha) \leq \epsilon\}$; comparisons at noise level ϵ are determined exactly up to $\text{Aut}_\epsilon(Q)$.*

Proof. If $f, g \in \text{Iso}(Q, Q')$ then $g^{-1}f \in \text{Aut}(Q)$ and $f = g(g^{-1}f)$; conversely $g\alpha \in \text{Iso}(Q, Q')$ for any $\alpha \in \text{Aut}(Q)$. The ϵ -statement follows by the same algebra with controlled distortion. $\square$

The theorem does three jobs. First, it gives the inverted spectrum its exact logical form: inversion is possible precisely when $\text{Aut}_\epsilon(Q)$ is nontrivial, and the space of possible inversions is that group. Second, it explains where inversion intuitions come from: projections of experience onto symmetric substructures. The bare hue circle has approximate rotational and reflective symmetry, so hue-only inversion is readily conceivable; the full ecological color structure, enriched by valence, memory color, and cross-modal association, breaks those symmetries, which is why a fully detailed inversion resists imagination. Third, it creates an experimental program: $\text{Aut}_\epsilon(Q)$ is estimable. Align a subject's empirical similarity structure with itself by Gromov–Wasserstein methods and count the nonidentity low-distortion couplings; the multiplicity and stability of alignments across subjects [28] directly measure the residual indeterminacy of inter-subjective comparison. As the measured structure is enriched, Aut_ϵ shrinks; we call the limiting claim *asymptotic structuralism*: every question about quality becomes empirical except the occupancy question already isolated by Theorem 1.

5.5 Temporal thickness

The experienced present has width and internal structure. On CCT both are set by the closed grain's dynamics: the width of the now is the grain's closure timescale τ , the horizon over which its macro-state screens off its micro-past, and the retention–protention structure of the now, in Husserl's sense, is the grain's own predictive–postdictive kernel: the current macro-state carries a compressed past and a committed expectation as constitutive components. This yields a measurable claim: individual differences in psychophysical integration windows (postdictive effects, audiovisual recalibration) should covary with individual differences in intrinsic neural timescales and with τ estimated from closure analysis. Section 7 develops the test.

5.6 The CCT theses

- T1.** (Substrate) A candidate bearer of experience is a closed grain: a Pareto-maximal coarse-graining with low interventional closure deficiency, nontrivial intrinsic computation, and an interface.
- T2.** (Subject) A subject is a closed grain implementing a centered self-model with presence tagging.
- T3.** (Quality) The character of an experience is its position in the grain’s intrinsic quality geometry, enriched by valence and action.
- T4.** (Center) Which grain is occupied is a token-indexical fact: real, but provably not derivable from structure (Theorem 1).
- T5.** (Time) The temporal grain of experience is the closure timescale of the substrate.
- T6.** (Epistemology) Claims under T1–T3 are adjudicated by interventional evidence and epistemic iteration; observational equivalence is constitutively insufficient.

6 What the theory explains or dissolves

6.1 The audit revisited

Each Section 2 problem receives a definite treatment. The explanandum problem: CCT distinguishes four explananda (substrate, subjecthood, quality, occupancy) and says which theories target which. The boundary and level problems: solved by the closure landscape; multiple maxima are permitted, so nested and dissociated subjects are possible rather than paradoxical. Observer-relativity: answered by intrinsic computation within closed grains, plus Corollary 2’s concession that one fact, occupancy, is indexical rather than structural. The quality problem: Theorem 2 plus the alignment program. Temporality: Section 5.5. Composition: two closed grains compose into one subject when their union admits a closed grain with a single centered self-model whose closure deficiency is below that of the parts taken separately; the sheaf-like question of when local centers glue into a global center is developed as an open problem in Section 10.

6.2 Deriving the meta-problem

A theory should explain the intuitions that generate the field [14]. CCT does so from its own resources.

(i) *The gap judgment.* A subject is a grain running equivariant inference over world-models. By Corollary 1, such inference can represent every structural fact and still not derive the occupancy of its own center. From inside, this presents as an unbridgeable residue: all the physical facts could be as they are, and the question “why is this one me, why is there something for me at all” remains open. The judgment is correct. What is incorrect is the further inference that the residue must be a missing *mechanism*. It is a missing *index*, and Theorem 1 proves no mechanism could supply it.

(ii) *Zombie conceivability.* A zombie world is a decentered duplicate: same object of $\mathcal{W}$, no occupied point. Since $U : \mathcal{W}_\bullet \rightarrow \mathcal{W}$ forgets exactly the basepoint, decentered descriptions are always consistent, which is why zombies are conceivable to any describer working in $\mathcal{W}$. Whether they are metaphysically possible is the question whether occupancy supervenes on structure, and

CCT deliberately brackets it: both the Russellian reading (occupancy as the intrinsic nature structure leaves open [51, 23]) and a deflationary indexical reading are compatible with every empirical commitment of the theory.

(iii) *Ineffability*. The self-model ρ compresses. The vehicles of its own tokens are not represented within it, so from inside, qualities present as primitive and simple even though, from outside, they are positions in a rich geometry. Ineffability is representational opacity at the fixed point of self-modeling, a phenomenon whose formal skeleton is the diagonal structure common to the classical limitative theorems [32].

CCT thus passes the meta-problem test while remaining realist: the puzzling judgments track a true structural fact, the non-derivability of the center, rather than an illusion [20].

6.3 Inverting the unfolding argument

Say two systems are *o-equivalent* if they induce the same distribution over input–output histories, and *i-equivalent* if there is an isomorphism of their interventional families over a common variable interface: every internal intervention on one has a counterpart with matching effects on the other. Unfolding a recurrent system into a feedforward equivalent preserves o-equivalence and destroys i-equivalence: the unfolded system has no internal grain whose perturbation has persistent, state-carrying effects.

Proposition 1 (Completeness of the interventional level). *Theories whose commitments distinguish systems only up to o-equivalence are undecidable by any evidence. Theories whose commitments are stated at the level of i-equivalence classes of closed grains are decidable in principle by perturbational evidence, and the unfolding construction, far from refuting them, certifies that perturbational evidence is the minimal sufficient currency.*

The empirical face of this proposition already exists: perturbational complexity measures [36, 11] are exactly rung-two evidence in Pearl’s hierarchy [43], and their diagnostic success is, on CCT, no accident. The recurrent-processing tradition [30] is vindicated in method even where its specific claims remain open.

7 Empirical program

We describe experiments in enough detail to expose the theory to risk. Throughout, “closure analysis” means estimating C_{pred} and proxies of C_{do} over a family of coarse-grainings of recorded or perturbed neural data.

E1. Automorphism estimation of quality spaces. Collect dense similarity judgments in several modalities (color, timbre, tactile texture, emotion terms) within single subjects. Estimate $\text{Aut}_\epsilon(Q)$ by Gromov–Wasserstein self-alignment: count stable nonidentity couplings as a function of ϵ and of structural enrichment (adding valence ratings, memory-color priors, cross-modal matches). Predictions: the bare hue circle yields a nontrivial Aut_ϵ containing approximate rotations and the red–green reflection; each enrichment strictly shrinks it; the fully enriched color structure is ϵ -rigid at achievable noise levels. Cross-subject alignment ambiguity [28] should match within-subject Aut_ϵ , since Theorem 2 says the former is controlled by the latter.

E2. Holonomy of quality. Diachronic comparison of one subject’s qualities is parallel transport along a path of states. Adaptation cycles (opponent-color fatigue, prism adaptation and removal, pharmacological on–off) form loops. Measure whether similarity structure returns

to identity around loops; nonzero loop displacement is quality holonomy, an empirical signature that diachronic quality identity is path-dependent.

E3. Closure landscapes across global states. Apply closure analysis to intracranial recordings (for example, the multi-state macaque ECoG of the Neurotycho corpus) across wakefulness, anesthesia, and sleep. Predictions: in wakefulness the landscape has a pronounced macro-scale closure optimum at timescales of order 10^2 – 10^3 ms; under propofol the optimum collapses (macro-dynamics lose autonomy) in register with perturbational complexity; under ketamine, closure is preserved while interface rank drops, matching preserved PCI and disconnected phenomenology.

E4. Perturbational separation of o-equivalent systems. Construct recurrent and unfolded feedforward networks matched to identical input–output behavior; verify observational indistinguishability and demonstrate interventional divergence (persistent versus bounded perturbation responses). This calibrates the metrics of Proposition 1 before their neural deployment.

E5. Split-brain closure bimodality. In callosotomy datasets, compute the closure landscape of each hemisphere and of the union. Prediction: complete section yields two closure optima whose union is not closed; partial section yields metastable alternation between one-optimum and two-optima regimes. The controversy over split-brain unity [45] is, on CCT, a signature of near-criticality of closure merging, and trial-by-trial dual-report incidence should track a momentary bimodality index.

E6. Temporal windows. Estimate each participant’s closure timescale τ from resting neurodynamics and, separately, their psychophysical integration windows (color-phi, cutaneous rabbit, audiovisual recalibration). T5 predicts covariation.

E7. Pharmacological geometry. Under psilocybin, collect within-session similarity judgments. Prediction: $\text{Aut}_\epsilon(Q)$ volume increases dose-dependently (near-symmetries proliferate as the geometry softens), correlating with reported boundary dissolution, and the entropic-brain observations [10] are reinterpreted as closure loosening.

E8. Brain-to-brain closure merging. In bidirectionally coupled dyads (as pursued in brain-to-brain interface programs), increase coupling bandwidth parametrically. CCT supplies an order parameter: the ratio of the dyad’s joint closure to the parts’ separate closures. Prediction: above a bandwidth threshold the landscape’s optimum shifts from two grains to one, and phenomenological unity reports track the shift. This turns the question “when do two subjects become one” from metaphor into measurement.

E9. Development and disorders. Closure emergence curves in infancy; closure with gated interface in dreaming (predicting unified but insulated phenomenology); presence-tag dissociation in depersonalization, where content processing is intact while tagging fails, predicting a specific interventional signature in tag-implicated circuitry.

Falsification conditions. F1: no coarse-graining of cortical dynamics shows a wake-specific closure optimum that collapses under anesthesia (kills T1’s neural relevance). F2: interventional and observational equivalence never dissociate in cortex, i.e., unfolded surrogates reproduce perturbation responses (kills T6 and much else). F3: Aut_ϵ structure fails to predict the modality profile of inversion intuitions (kills the Section 5.4 account of quality). F4: psychophysical integration windows are uncorrelated with intrinsic timescales and closure τ (kills T5). F5: presence-tag interventions dissociate from presence phenomenology (kills T2’s tagging component).

8 Artificial systems

Lemma 1 (Mimicry). *Fix any behavioral test: a protocol whose verdict is a function of an interaction transcript of bounded length over a finite alphabet. For any system that passes it, there exists an o-equivalent system, realizable as a finite lookup structure, that also passes it and whose every supra-gate grain has trivial closure and no centered self-model.*

Proof. Tabulate the passing system’s conditional output distribution over the finite transcript tree and realize it feedforward. o-equivalence is by construction; the realization has no internal state-carrying grain above the gate level. $\square$

Behavioral certification of machine consciousness is therefore impossible in principle, however long the conversation. This does not license agnosticism; it redirects assessment to the properties CCT says matter, all of which are white-box:

(i) *Closure analysis of the running system.* Does the system, at deployment, admit macroscopic grains with low interventional closure deficiency over behaviorally relevant timescales? For present transformer architectures the token-level computation is an unfolded pass, and CCT accordingly predicts thin closure at that grain despite rich competence: competence without closure. Persistent-memory agentic loops change the analysis; the relevant unit is the agent process, and closure can emerge at that level whether or not anyone intends it.

(ii) *Centered self-model probes.* Interpretability methods can ask whether internal variables implement a self-token with presence tagging carrying the right interventional profile, rather than whether the system *says* so.

(iii) *Quality-geometry richness.* The intrinsic similarity geometry of internal states under intervention is measurable and comparable across systems and to biological structures by the alignment methods of Section 5.4.

The practical proposal is a scorecard along these axes replacing behavioral tests, which upgrades indicator-based assessment [8] from theory-borrowed checklists to measurements with a common formal spine. Because closure can emerge unintended in agentic deployments, monitoring it is also a welfare precaution.

9 Relations to existing theories

IIT [41, 1]: CCT shares the intrinsicness ambition and the substrate-first orientation, replaces Φ -maximization and exclusion with the closure landscape (computable, multi-optimum, interventionally defined), and relocates IIT’s identity claim: quality is geometry of closed dynamics rather than of cause–effect repertoires. GNWT [3, 35]: the workspace is, on CCT, the access architecture by which contents get bound under the presence tag into the centered model; CCT adds the substrate and quality layers that workspace theory brackets. Higher-order and reality-monitoring theories [48, 31]: absorbed as the functional side of presence tagging, with truth-conditions given interventionally rather than representationally. Attention schema [24] and self-model theory [38]: precursors of the centered self-model, now anchored to a substrate criterion. Free energy [21]: the viability field V echoes its concerns; closure replaces blankets where individuation is at stake. Recurrent processing [30]: vindicated methodologically by Proposition 1. Russellian monism [51, 23]: CCT is compatible with it and does not require it; the theory’s empirical content is invariant across the Russellian and deflationary readings of occupancy. Illusionism [20, 18]: CCT matches its meta-problem discipline while diagnosing the judgments as truth-tracking. The intrinsic-computation lineage [17, 53] supplies the substrate mathematics, and control-theoretic

treatments of workspace architecture suggest natural implementations of the interface operator through balanced input–output modes.

10 Objections and open problems

Is centering a solution or a label? The charge: T4 renames the hard problem. Reply: labels do not prove theorems. CCT shows the residue is exactly a basepoint, shows no structural rule can supply it, derives the field’s intuitions from that fact, and renders every remaining question empirical. Whether one calls the residue solved or delimited, it is no longer vague.

Arbitrariness of the closure functional. Choices of divergence, lattice, and thresholds could make closure as conventional as Φ . This is the most serious internal risk; Version 2 must supply invariance results and worked numerics.

Panpsychist slippage. Closure is graded, so many systems have a little of it. CCT bites the gradation bullet for substrates while reserving subjecthood for grains with centered self-models, and it says which axes the grading runs along.

The gluing problem. When do overlapping closed grains with local centers compose a single subject? We have a criterion sketch (Section 6.1) and no general theory; the obstruction-theoretic formulation is open.

De se skepticism. Some deny that indexicality is philosophically deep [9]. Even on deflationary views, the derivability asymmetry captured by Theorem 1 stands, and it is all the theory needs.

11 Conclusion

The paper’s wager is that consciousness science has been trying to do three impossible things at once: derive a centered fact from uncentered theory, individuate bearers without a criterion, and validate substrate claims with observational data. CCT separates the strands. It proves what cannot be done and builds instruments for the rest. The measure of the proposal is Section 7: a theory of consciousness should be judged by the experiments it makes possible and the ways it permits itself to fail.

A Problem catalogue

P1 hard problem and explanatory gap [12, 33]. P2 measurement circle and report confound [2]. P3 access–phenomenal dissociation [6]. P4 theory incommensurability exposed by adversarial collaboration [16]. P5 unfolding underdetermination [19]. P6 boundary problem. P7 level and grain problem [27]. P8 combination and decomposition. P9 observer-relativity of computation [46, 52, 13]. P10 Newman underdetermination of structure [40]. P11 quality determination and inversion. P12 inter-subject comparison [28]. P13 temporal thickness and postdiction [26]. P14 causal role and epiphenomenalism. P15 meta-problem [14]. P16 AI assessment [8].

B Idea inventory

Verdicts: **A** adopted as load-bearing, **M** adopted with modification, **I** adopted as interpretation or precedent, **D** deferred, **R** rejected.

1. Yoneda and enriched categories: identity through relational profile, up to isomorphism; the “up to isomorphism” is exactly the gap Theorem 2 measures. **A**. 2. Sheaves and obstruction

theory: binding and combination as gluing with cohomological obstructions. **M** (sketch only; open problem). 3. Topos internal logic: the first-person as the internal language of a structured object; phenomenal indeterminacy as intuitionistic failure of excluded middle. **I**. 4. Lawvere’s fixed-point unification of the limitative theorems: self-model opacity and ineffability. **A** (Section 6.2). 5. Gauge theory: quality instantiation as gauge freedom; inter-subject comparison as choice of transport; holonomy as measurable path-dependence. **M** (as mathematics of description, with no physics identification claimed). 6. Klein’s Erlangen program: a science of consciousness as the study of invariants under the group of re-descriptions. **A** (methodological spine of T4, T6). 7. Löwenheim–Skolem and Newman: structure never fixes reference; hence a non-structural primitive is mandatory. **A**. 8. Pointed categories and de se content: the basepoint as primitive. **A** (core). 9. Renormalization and coarse-graining: level selection by flow; conscious level as a distinguished scale. **M** (via closure landscape rather than RG fixed points). 10. Lumpability and causal abstraction: closure formalization. **A** (core). 11. Computational mechanics and ε -machines: intrinsic computation; the answer to triviality. **A**. 12. Koopman operator theory: intrinsic observables as spectral objects. **D** (alternative formalization of closure; promising). 13. Causal emergence: macro causal power. **M** (closure is lawfulness, not effect-size maximization; dissociation testable). 14. Markov blankets: boundaries as conditional independence. **R** as individuator, retained as a special case [7]. 15. Interventionist causal inference: the epistemology; rung-two evidence. **A**. 16. Interactive proof and cryptographic framing: limits of black-box verification. **A** (Lemma 1). 17. Optimal transport and Gromov–Wasserstein alignment: the experimental engine for quality comparison. **A**. 18. Information geometry: metrics on state and belief spaces. **I** (background tool). 19. Stochastic thermodynamics: the arrow of felt time from entropy production. **D** (future work on T5). 20. Relational quantum mechanics and QBism: precedent that fundamental physics admits perspectival facts. **I** (precedent only). 21. Quantum Darwinism: objectivity as redundant records; by inversion, privacy as non-broadcast information within the closure boundary. **I**. 22. Objective-collapse and Orch-OR: quantum substrate claims. **R** (no independent evidence; grain mismatch). 23. Homotopy type theory and univalence: identity of equivalent structures; clarifies the semantics of inversion claims. **I**. 24. Persistent homology: empirical shape of quality spaces across scales. **A** (methods for E1, E7). 25. Whitehead’s occasions: experience as eventual. **I** (resonates with discrete-moment findings; not formalized here). 26. Husserl’s retention–protention: the target analysis of temporal thickness. **A** (operationalized in T5). 27. Nishida’s logic of place and the Buddhist analysis of non-self: the center as place rather than ego-substance. **I** (developed in Version 2). 28. Chang’s epistemic iteration: measurement without ground truth as a bootstrap. **A** (T6). 29. Autopoiesis and closure to efficient causation: the organizational-closure lineage behind T1. **I** (ancestry acknowledged). 30. Illusionism and the meta-problem: the adequacy condition that intuitions be derived. **A** (as constraint, with realist verdict).

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